

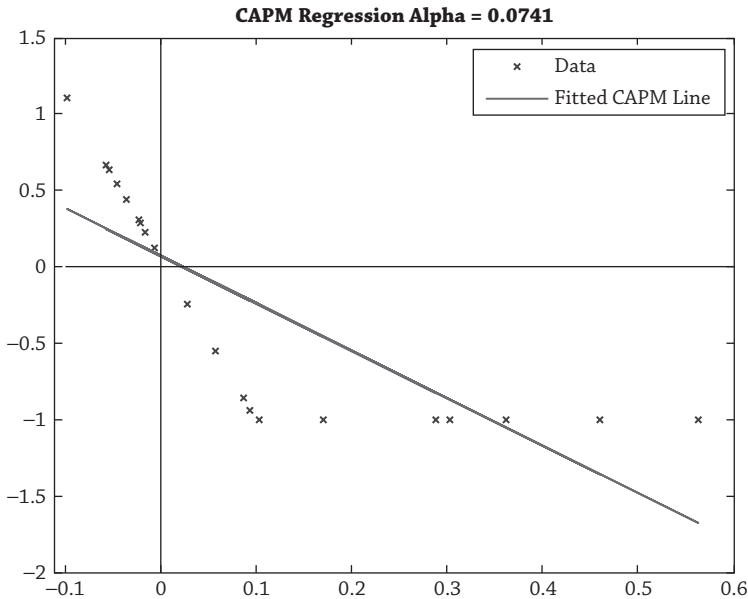


Figure 10.7

3.4 NON-LINEAR PAYOFFS

With alphas and information ratios, any manager can appear to have talent when he actually doesn't.

Alphas are computed in a *linear* framework. There are many *nonlinear* strategies, especially those involving dynamic option strategies, that can masquerade as alpha.¹⁵ To give an extreme (and admittedly stylized) example, consider Figure 10.7. It is produced by selling put options on the market portfolio in a small sample, just using Black–Scholes (1973) prices. The returns on these put options are recorded with crosses. I then run a CAPM regression with these simulated returns. The “alpha” appears to be positive—ta da!—but we know that in the Black–Scholes world there is no extra value created in puts or calls. The alpha is purely illusory. This is not a result of a small sample, even though small samples exacerbate the problem. *No nonlinear strategy can be adequately captured in a linear framework.*¹⁶ This is a serious problem because many common hedge fund strategies, including merger arbitrage, pairs trading, and convertible bond arbitrage, have payoffs that resemble nonlinear option volatility strategies.¹⁷

¹⁵This was first shown in a seminal paper by Dybvig and Ingersoll (1982). Technically, this is because any factor model used (Dybvig and Ingersoll used the CAPM) implicitly allows the pricing kernel to be negative and permits arbitrage (see chapter 6). The linear pricing kernel correctly prices all benchmark assets (stocks) but incorrectly prices nonlinear payoffs like derivatives.

¹⁶For a more formal treatment see Lhabitant (2000), Lo (2001), and Guasoni, Huberman, and Wang (2011). You can *always* beat a market Sharpe ratio (or information ratio) by selling volatility.

¹⁷See Mitchell and Pulvino (2001), Gatev, Goetzmann, and Rouwenhorst (2006), and Agarwal et al. (2011), respectively.